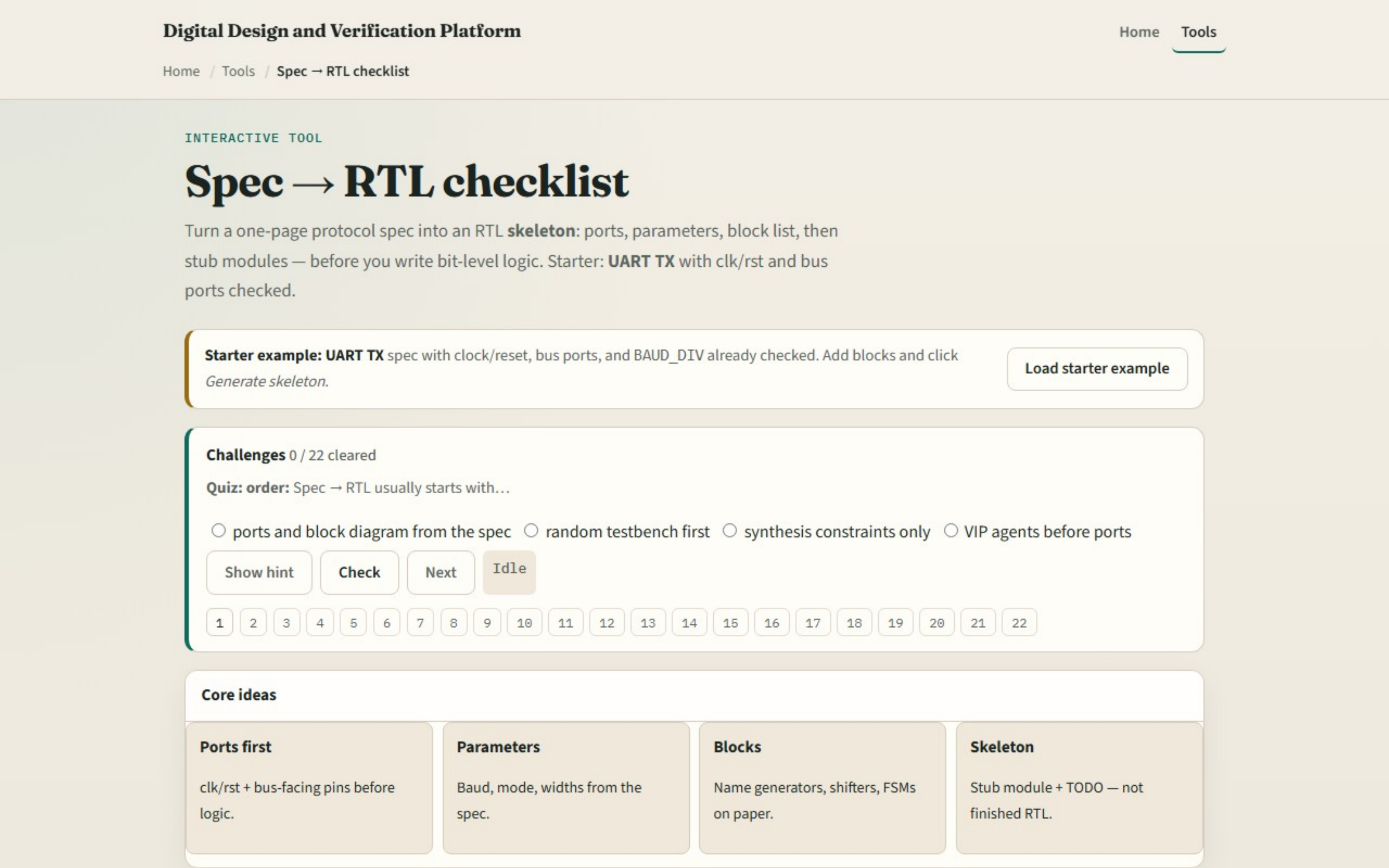
Spec to RTL

Before you write bit-level logic, translate the spec into structure

UART TX starter

- Starter preset: UART transmit eight-N-one
- Tx busy, plus baud parameter BAUD_DIV
- Required blocks still waiting
- Check the block rows, then click Generate skeleton to grow a uart_tx module outline

Browser lab



Spec to RTL starter

Real RTL/TB practice

- In Track A, from the UART TX spec list every port name and one parameter
- Name three blocks you would draw on paper before coding
- Optional: compare your list to a UART sketch in the legacy combined materials
- Write one sentence on what belongs in the skeleton versus what belongs in the FSM

Pitfalls to watch

- Do not skip ports and jump straight to the FSM
- Parameters are not optional when baud or mode defines behavior
- A skeleton with TODO is not done RTL, this lab separates planning from implementation
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish required UART checks and generate the skeleton once
- On paper, list ports, parameters, and three blocks for UART TX
- When you are ready, take the short quiz, then continue to baud divider